Maaz Qureshi

Research Engineer (ML, Embodied AI and Robotics): MASc Mechtronics University of Waterloo

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SKILLS

Machine Learning & Foundation Models: Vision-Language-Action models (π_0, smolVLA), World-Action Models (literature), Parameter-Efficient Fine-Tuning (PEFT: LoRA), CNNs, VLMs, Quantization (FP16/INT8), Reinforcement Learning (PPO, policy optimization), Imitation Learning (behavior cloning) & Diffusion Policy

Robotics: 6D Pose Estimation, Visual-Inertial SLAM (volumetric mapping & localization), Motion Planning for 7-DoF Manipulators (IK/FK), Sensor Fusion (stereo RGB-D, LiDAR, mmWave 4D radar, IMU), 5G-Connected Factory Robotics, AR/XR (ARKit)

Languages: Python, C++, PyTorch, TensorRT (CUDA), Eigen, NumPy, ROS2

Tools: Docker, CI/CD, gRPC, NVIDIA Isaac, MuJoCo, Gazebo, RViz, Asynchronous GPU Pipeline, Debugging/Monitoring (Valgrind, Jtop)

EXPERIENCE

Embodied AI Robotics RA (Research Assistant) May 2026 – Present

Apple Inc. (joint research with University of Waterloo) *Waterloo, Canada*

- Confidential NDA project with Apple Robotics and University of Waterloo for 7-DoF manipulator motion planning.
- Fine-tuning/training the VLA model(s) and diffusion policy(s) for the pick and place tasks (confidential).
- Built an iPhone Pro (ARKit)-based egocentric human-demonstration data pipeline for PEFT (LoRA-based) fine-tuning of foundation models.
- Characterized ARKit pose-tracking accuracy against Franka Panda 6D pose and gripper-width ground truth across a 5–65°/s angular-velocity sweep, identifying a sub-40°/s operating envelope (mean translational error $<$10%) for reliable demonstration collection.
- Developing a UMI-style handheld gripper with a custom Franka-flange mount for scalable demonstration collection and deployment.

Robotics Software Engineer [video] September 2025 – April 2026

Cobionix Corporation Inc. *Kitchener, Canada*

- Developed a 22 FPS real-time multimodal RGB-D ML pipeline (SmolVLM, pose estimation, object detection, Segmentaion) in C++/CUDA.
- Applied PEFT (LoRA) to fine-tune the SmolVLM vision-language backbone for object recognition.
- Applied TensorRT FP16/INT8 quantization in async GPU pipelines on NVIDIA Jetson to boost throughput and cut latency.
- Implemented real-time RGB-D stereo sensor fusion with intrinsic/extrinsic spatiotemporal calibration ($>$90% alignment accuracy).

Graduate Research Student: ML Robot Perception & Manipulation January 2024 – August 2025

University of Waterloo, RoboHub Lab [Publication, Video, GitHub] *Waterloo, Canada*

- Developed a distributed multi-robot dense volumetric SLAM system (C++/ROS2 Humble) with TSDF-octree mapping and ORB-SLAM3.
- Achieved an 85% improvement in map coverage over the state-of-the-art baseline across 4 autonomous mobile robots (AMRs).
- Fused mmWave 4D radar with RGB-D sensing for perception in cluttered, low-visibility settings (92% detection accuracy).

MITACS Industrial Research Internships:

Keysight Technologies Inc. [Publication, Video, GitHub] *Thesis Internship*

- (SIM-to-REAL) Developed an OMPL/RRT* sampling based motion-planning pipeline for 7-DoF Franka and KUKA collaborative robots.
- Benchmarked a PPO reinforcement-learning planner against RRT sampling-based motion planning for manipulator trajectories.
- Validated sim-to-real deployment, achieving 100% joint-limit and collision avoidance across hemispherical scan patterns.
- Built with Python, ROS/RViz, and Docker for end-to-end pipeline development and deployment.

Rogers Communications Inc. *Thesis Internship*

- Integrated 5G connectivity into a distributed software pipeline on NVIDIA GPUs for smart-factory robots, achieving a 37% latency reduction.
- Benchmarked 5G sub-6 GHz (3.5 GHz) and mmWave (28 GHz) channels against WiFi for industrial robotics applications.

Asset Integrity Engineer: (Robotics & ML) May 2022 – December 2023

ABYSS Solutions Inc. *Islamabad, Pakistan (Onsite) & Remote Australia*

- Performed anomaly outlier detection (supervised ML, U-Net segmentation) for corrosion detection on oil/gas platforms.
- Performed offline SLAM mapping visual model for mooring-chain and dam inspection using underwater ROV imagery.[Link]
- Led a 10-engineer team delivering end-to-end digital inspection of 18,000+ assets (deadlegs inspection) with 91% high-quality coverage.
- Actively involved in special projects of organixation for UAV, Virtual reality and Underwater robotics ops.

EDUCATION

University of Waterloo — Waterloo, Canada
Master of Applied Science (MASc), Mechatronics Engineering ***(CGPA: 93.5%)*** — *January 2024 – August 2025*

- **Thesis link:** *"Intelligent Multi-Robot Autonomy with Connected AMRs and Manipulators for SMART Factory(s)"*, Transcript, Degree
- **Advisor(s):** Dr. William Melek and Dr. George Shaker . Defence committee: Dr. Soo Jeon, and Dr. Gennaro Notomista

Bahria University — Islamabad, Pakistan
Bachelor of Science (BSc), Electrical (Electronics) Engineering ***(CGPA: 80.5%)*** — *September 2017 – August 2021*

- **Thesis link,** Video: *"Swarm Autobots"*, Transcript, Degree
- **Advisor:** Dr. Saleem Aslam Defense committee: Dr. Adil Raja

SIDE PROJECTS

- **Diffusion Policy vs Imitation learning in Navigation:** [Link] benchmarked diffusion-policy, behavior-cloning, and LSTM imitation learning for TurtleBot 4 navigation; added YOLOv11 for real-time safety detection.
- **Visual Feature Extraction with Machine Learning:** Applied KNN and SVM regression models with FRIQUE image-quality metrics to benchmark visual feature extraction on computer-generated vs. real-world image datasets.
- **Humanoid Robotics: Non-Verbal Human-Robot Interaction** [Link]: Designed a non-verbal gesture-recognition model for multimodal control of a NAO humanoid robot assisting in collaborative breakfast preparation.
- **AMRs in Health Care** [Link]: Built an RFID-based autonomous nursing robot for pill dispensing, real-time perception, and live video streaming in healthcare environments.
- **UAS Challenge 2022, IMechE UK: Aerial Robotics** [Link]: Team Supervisor for a hexacopter UAV (6.9 kg takeoff weight, 100 m operating altitude) for locust-swarm spraying, with a custom carbon-fiber S950 frame and 340 KV BLDC motors delivering 2920 g of thrust.

SELECTED PUBLICATIONS:

- **[C1] M. Qureshi et al.**, "CRADMap: Applied Distributed Volumetric Mapping with 5G-Connected Multi-Robots and 4D Radar Perception," *IEEE ICARM 2025.* **(Published)** [IEEE Xplore]
- **[C2] M. Qureshi et al.**, "Hemispherical Angular Power Mapping of Installed mmWave Radar Modules Under Realistic Deployment Constraints," *IEEE APS 2026.* **(Published)** [IEEE Xplore]
- **[J1] M. Qureshi et al.**, "RAPTAR: Radar Radiation Pattern Acquisition through Automated Collaborative Robotics," *IEEE Robotics and Automation Letters (RA-L).* **(Under Revision)** [arXiv]

HONORS & AWARDS

- **Full Graduate Funding Award**, Mitacs Canada & University of Waterloo (MASc) — *2024–2025*
- **Top 3 Undergraduate Thesis Project Prize**, Bahria University Video — *2021*
- **Star of the Month (Best Performer) Award**, Abyss Solutions Inc. [Link] — *March 2023*
- **3× Hackathon Winner**, Abyss Solutions Robotics & VR Innovation Challenges — *2022–2023*

OTHER EXPERIENCE

TA, University of Waterloo — *Spring 2024 – Spring 2025*
ME 123 (Electrical Eng. for Mechanical Engineers), Spring 2024 | ME 360 (Automatic Control Systems), Fall 2024 & Winter 2025 | MTE (Advanced Calculus), Spring 2025

Conference Presentation, IEEE APS 2026 — *Jul. 2026*
Detroit, Michigan, USA

Conference Presentation, IEEE ICARM 2025 — *Aug. 2025*
Portsmouth, United Kingdom

Symposium Presentation: RoboHub Symposium, Poster & Research Talk — *Jan. 2025*
University of Waterloo, Canada

Conference Presentation, Rogers "Beyond Connectivity" Summit — *Jul. 2024*
Brampton, Canada

PEER REVIEW SERVICE

- **Reviewer, IEEE International Conference on Robotics and Automation (ICRA) 2026** (Vienna, Austria; Submission #1761): *"ROR-SLAM: UWB Radar-based Object Recognition for Feature-based SLAM."*
- **Reviewer, IEEE Sensors Journal** (Manuscript Sensors-69555-2024): *"EAMT-SLAM: Edge-Aided Monocular Thermal Sensor SLAM in Low-Illumination Environments."*
- **Reviewer, MEIE 2026** (9th Int'l Conference on Mechanical, Electric, and Industrial Engineering; Manuscript MEIE99621): *"Feasibility Analysis and Reinforcement Design of a Portable Radar for Use in Vehicle Environments."* Recommended minor revisions.